

Affine versus permutation equivalence of cubic monomial rotation symmetric Boolean functions in 33 variables

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Abstract

We give an exact computational proof that two cubic monomial rotation symmetric Boolean functions in 33 variables are equivalent under an invertible affine input substitution if and only if they are equivalent under a coordinate permutation. Each function is the sum of the distinct cyclic translates of one squarefree cubic monomial, including the orbit of length eleven. Enumeration gives twelve multiplier classes. Hamming weights separate 64 of their 66 unordered pairs; a fourth Walsh moment modulo 8191 and the number of cubic-polar contractions of rank twelve separate the remaining two pairs. The contraction calculation covers all 2^{33} directions through checked cyclic orbits. We give the invariant identities, coverage argument, exact values, and reproducible programs. The conclusion is confined to dimension 33, and no priority claim is made.

1 Statement and context

Index the coordinates by $\mathbb{Z}/33\mathbb{Z}$. For a three-element subset S , let $\mathcal{E}(S) = \{S + t : t \in \mathbb{Z}/33\mathbb{Z}\}$ be a set of distinct triples, and put

$$f_S(x) = \sum_{E \in \mathcal{E}(S)} \prod_{i \in E} x_i \quad \text{in} \quad B_{33} = \mathbb{F}_2[x_0, \dots, x_{32}] / (x_i^2 - x_i).$$

The sum is in $\mathbb{F}_2$, with each distinct monomial included once. Write $\mathcal{C}_{33} = \{f_S : |S| = 3\}$ and abbreviate $f_{\{0,a,b\}}$ to $f_{0,a,b}$. Affine equivalence here means input substitution only: there is no additional affine output function or output complementation.

Theorem 1. *For $f, g \in \mathcal{C}_{33}$, if*

$$g(x) = f(Ax + b) \quad (x \in \mathbb{F}_2^{33})$$

for some $A \in \text{GL}(33, 2)$ and $b \in \mathbb{F}_2^{33}$, then $g(x) = f(Px)$ for a coordinate-permutation matrix P .

The converse is immediate. The theorem asserts the existence of a coordinate permutation realizing an equivalence; it does not describe every affine map that realizes one.

Cusick [1, Remark 3.9] conjectures this implication in arbitrary dimension for cubic monomial rotation symmetric functions. The inspected version reports verification through $n \leq 32$, using frequencies of absolute Walsh-spectrum values. Its Theorem 3.8 concerns the action of permutations preserving rotation symmetry. We do not use that restricted orbit theorem to infer a statement

about arbitrary affine substitutions. Instead, we enumerate the domain at $n = 33$ and separate explicit representatives by affine invariants.

Relative to the range reported in that source, the present contribution is the complete specified layer $n = 33$, with only two separations beyond Hamming weight. This paper does not settle the conjecture in arbitrary dimension. The literature comparison, as of September 8, 2026, comprises the versioned source and the three abstracts returned by an arXiv query for rotation symmetry and affine equivalence. Citing literature and other databases were not exhaustively screened. Thus we make no claim of priority or of the previously unsettled status of dimension 33. For broad adjacent context on exact integer certification of counts under permutation symmetry, see Zeng [2]; that reference concerns posets and supplies no Boolean-function classification used here.

2 Twelve representatives and their weights

Multiplication of indices by a unit $u \in (\mathbb{Z}/33\mathbb{Z})^\times$ is a coordinate permutation and sends $\mathcal{E}(S)$ to $\mathcal{E}(uS)$. There are twenty such units. We call their orbits on $\mathcal{C}_{33}$ *multiplier classes*.

Exact computation 1. *The $\binom{33}{3} = 5456$ triples generate 166 distinct functions and the twelve multiplier classes in Table 1. The table gives their exact Hamming weights. Only the pairs*

$$(f_{0,1,3}, f_{0,1,4}), \quad (f_{0,1,6}, f_{0,1,10})$$

have equal weights among different representatives.

Table 1: Complete multiplier-class enumeration at $n = 33$. Class size counts functions, not monomials.

Representative S	Class size	wt(f_S)
(0, 1, 2)	10	4219743232
(0, 1, 3)	20	4280188928
(0, 1, 4)	20	4280188928
(0, 1, 5)	20	4277820416
(0, 1, 6)	20	4278003712
(0, 1, 7)	20	4278702080
(0, 1, 10)	10	4278003712
(0, 1, 11)	10	4273810432
(0, 1, 12)	20	4273813504
(0, 3, 6)	5	4221368320
(0, 3, 9)	10	4273160192
(0, 11, 22)	1	4113568768

Here is a direct reproducible enumeration. Construct $\mathcal{E}(S)$ for every triple S and retain the distinct edge sets. From the unclassified edge sets choose the lexicographically smallest triple appearing in any of them, form all twenty unit images of its edge set, record that class, and remove it. Continue until none remain. The resulting class sizes sum to 166. Of these functions, 165 have 33 monomials and one has eleven; the identity $165 \cdot 33 + 11 = 5456$ also checks the triple count. The supplement reconstructs the classes from all triples without assuming a classification theorem.

For the weights, define the signed partition sum

$$Z(f) = \sum_{x \in \mathbb{F}_2^{33}} (-1)^{f(x)}, \quad \text{wt}(f) = \frac{2^{33} - Z(f)}{2}.$$

One exact algorithm attaches the factor $(-1)^{x_i x_j x_k}$ to each edge and eliminates variables by multiplying the factors containing that variable and summing over its two values. This preserves $Z(f)$. Factors summarize disjoint sets of previously eliminated variables, so intermediate magnitudes are at most 2^{33} .

The supplied second implementation instead uses a binary transfer matrix. For a full-orbit function $f_{0,a,b}$, its states are length- b words, and its nonzero transitions are

$$(s_0, \dots, s_{b-1}) \longrightarrow (s_1, \dots, s_{b-1}, c), \quad \text{weight } (-1)^{s_0 s_a c}.$$

The trace of its 33rd power is $Z(f_{0,a,b})$: closed walks with a fixed starting position correspond bijectively to indexed cyclic binary assignments. In particular there is no division or multiplication by 33. The trace is computed by propagating each boundary state separately. Every partial path sum, and the accumulated diagonal sum, has magnitude at most 2^{33} , so signed 64-bit arithmetic suffices.

For $S = (0, 3, 6)$ and $(0, 3, 9)$ the coordinates split into three disjoint 11-cycles; the relevant 11-variable partition sums are cubed. For $S = (0, 11, 22)$ there are eleven disjoint monomials, each with signed partition sum $7 - 1 = 6$, giving $Z = 6^{11}$. These formulas reproduce all entries of Table 1 and handle the short orbit explicitly. Since an invertible affine map permutes all inputs, Hamming weight separates 64 of the 66 pairs.

3 A fourth Walsh moment separates the first collision

For a Boolean function on $\mathbb{F}_2^n$, put $F(x) = (-1)^{f(x)}$ and

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} F(x)(-1)^{a \cdot x}, \quad K_4(f) = 2^{-n} \sum_{a \in \mathbb{F}_2^n} W_f(a)^4.$$

Lemma 1. $K_4(f)$ is an integer invariant under invertible affine input substitution, and

$$K_4(f) = \sum_{x, y, z \in \mathbb{F}_2^n} F(x)F(y)F(z)F(x + y + z). \quad (1)$$

Proof. For $g(x) = f(Ax + b)$, a change of variables gives

$$W_g(a) = (-1)^{(A^{-T}a) \cdot b} W_f(A^{-T}a).$$

Fourth powers remove the sign and A^{-T} permutes the frequencies. Expanding the fourth powers and summing over a gives 2^n precisely when the four inputs sum to zero, and zero otherwise. This proves (1), including integrality. $\square$

For $f_{0,1,r}$ with $r = 3, 4$, assign the eight-valued spin $s_i = (x_i, y_i, z_i) \in \mathbb{F}_2^3$ to coordinate i . The local exponent in (1) is

$$h(a, b, c) = \sum_{j=1}^3 a_j b_j c_j + \left(\sum_{j=1}^3 a_j \right) \left(\sum_{j=1}^3 b_j \right) \left(\sum_{j=1}^3 c_j \right) \quad \text{in } \mathbb{F}_2. \quad (2)$$

Let T_r be indexed by words of length r in these spins. Its nonzero transitions are

$$(a, s_1, \dots, s_{r-1}) \longrightarrow (s_1, \dots, s_{r-1}, c), \quad \text{weight } (-1)^{h(a, s_1, c)}.$$

The same closed-walk correspondence gives $K_4(f_{0,1,r}) = \text{tr}(T_r^{33})$. There are 8^r states, each with eight successors. Propagating each boundary vector for 33 steps modulo 8191, and summing its diagonal entries, yields the exact residues

$$K_4(f_{0,1,3}) \equiv 177, \quad K_4(f_{0,1,4}) \equiv 2106 \pmod{8191}. \quad (3)$$

Every entry is reduced to $\{0, \dots, 8190\}$ after a step; each signed sum before reduction has magnitude at most $8 \cdot 8190$. Thus this is integer modular arithmetic, with no floating-point approximation. The supplied verifier evaluates (2) literally. A separate implementation uses an eight-point Walsh transform to perform the local sums. The two implementations agree. The unequal residues in (3) prove that the integer invariants differ and exclude the first equal-weight pair.

4 Cubic-polar contraction ranks separate the second collision

Write $D_u f(x) = f(x+u) + f(x)$ in $\mathbb{F}_2$. For a cubic Boolean function define

$$\tau_f(u, v, w) = D_u D_v D_w f(0).$$

The third derivative is independent of the base point. Expanding a squarefree cubic monomial gives the six assignments of the three directions to its three coordinates. Terms of degree at most two vanish. Consequently τ_f is an alternating trilinear form.

For each u define the alternating matrix $B_f(u)$ by $v^T B_f(u) w = \tau_f(u, v, w)$, and put

$$N_{12}(f) = \#\{u \in \mathbb{F}_2^{33} : \text{rank}_{\mathbb{F}_2} B_f(u) = 12\}.$$

Lemma 2. *N_{12} is invariant under invertible affine input substitution.*

Proof. The derivative identity for $g(x) = f(Ax + b)$ is $\tau_g(u, v, w) = \tau_f(Au, Av, Aw)$, because the third derivative of f does not depend on its base point. Hence

$$B_g(u) = A^T B_f(Au) A. \quad (4)$$

Congruence by A preserves rank, and $u \mapsto Au$ permutes all directions. This proves the claim, including translations. $\square$

Each edge $\{i, j, k\}$ contributes u_i to both entries (j, k) and (k, j) , with the analogous contributions from u_j and u_k . This gives a direct binary construction of $B_f(u)$ from the distinct edges. For $f_{0,1,r}$, a set bit u_k equivalently toggles the symmetric pairs

$$(k+1, k+r), \quad (k-1, k+r-1), \quad (k-r, k-r+1),$$

with indices modulo 33.

The complete exact calculation gives

$$N_{12}(f_{0,1,6}) = 3058, \quad N_{12}(f_{0,1,10}) = 2959. \quad (5)$$

For clarity about its coverage, cyclic rotation of u conjugates $B_f(u)$ by a coordinate rotation. Rank can therefore be computed once per binary necklace and credited by the necklace’s least period. The completed enumeration has the following period counts:

least period p	1	3	11	33
necklaces	2	2	186	260300986.

There are 260301176 representatives in total, and

$$2 \cdot 1 + 2 \cdot 3 + 186 \cdot 11 + 260300986 \cdot 33 = 8589934592 = 2^{33}. \quad (6)$$

The verifier starts with the all-zero word. To obtain its next candidate, it changes the rightmost zero to one and repeats the resulting prefix to fill the word. If the prefix length divides 33, it tests that candidate. For every emitted word it checks strict increase in binary value, minimality among all rotations, and the actual least period by rotation. Strict increase and minimality imply that the credited orbits are disjoint; the verified period is their size. Equation (6) then establishes completeness, without requiring a correctness theorem for the candidate generator.

The supplied verifier constructs the contraction tensor from the edge set, precomputes sums for blocks of eleven direction bits, and computes ranks using alternating elimination. If $B_{ij} = 1$, remove its invertible 2×2 alternating block; on the remaining indices the Schur complement is

$$B'_{k\ell} = B_{k\ell} + B_{ki}B_{j\ell} + B_{kj}B_{i\ell}.$$

This contributes two to the rank. Isolated zero rows contribute zero. Stopping after seven such blocks certifies rank greater than twelve. The output bucket labelled 14 means only “rank greater than 12”; no exact claim for higher ranks is needed. The low-rank counts are

rank	0	2	4	6	8	10	12
$f_{0,1,6}$	1	0	0	33	132	528	3058
$f_{0,1,10}$	1	0	0	33	132	528	2959.

An implementation using ordinary binary Gaussian elimination reproduces these counts in a separate exhaustive run. Positive rank-twelve representatives were also checked individually; completeness rests on the full enumeration, not on those positive witnesses alone.

Proof of Theorem 1. Every member of $\mathcal{C}_{33}$ is related by an explicit coordinate permutation to a representative in Table 1. Different representatives have different weights, except for two pairs. Those pairs are separated by (3) and (5), respectively, using Lemmas 1 and 2. Thus all 66 cross-class pairs are affine inequivalent. An affine equivalence between any two members, composed with their known coordinate permutations, induces an affine equivalence between their representatives. The representatives must coincide. The two members therefore lie in the same multiplier class, which supplies the required coordinate permutation. $\square$

5 Reproduction and computational scope

The supplement contains the C++17 verifier `verify.cpp`, a Python 3 driver `reproduce.py`, exact expected values, and replay instructions. From its directory, the single command

```
python3 reproduce.py
```

compiles the verifier with assertions enabled, enumerates all triples and multiplier classes, recomputes all twelve weights and both Walsh residues, and performs the complete contraction enumeration. It writes separate replay outputs. Its C++ modes may also be run directly:

```
clang++ -O3 -std=c++17 verify.cpp -o verify
./verify walsh 33 3 2
./verify walsh 33 4 2
./verify polar 33 6 10
```

The final argument in Walsh mode is an unused interface placeholder. Do not compile with `-DNDEBUG`, which disables coverage assertions.

Verification accompanying the computation used separately written implementations of the scalar transfer sum, the local fourth-moment transition, matrix construction, rank elimination, and necklace generation. The scalar transfer algorithm was checked against complete truth tables for all 30 cyclic functions at $3 \leq n \leq 9$. The fourth-moment verifier agreed with 21 direct Fourier calculations for $r = 3, 4$ and $r + 1 \leq n \leq 14$. Binary contraction ranks agreed with dense Gaussian elimination in nine smaller cases, totaling 16352 function–direction evaluations. At $n = 33$, all 185 recorded representative/function pairs of rank twelve were individually verified, with period-weighted counts 3058 and 2959. These small checks supplement the completed dimension-33 computations; they do not substitute for them.

The finite numerical assertions depend on the exact programs and their exhaustive completion. No sampling or floating-point evidence enters the proof. The computation does not claim a full Walsh-spectrum classification, a classification of all affine automorphisms, or a result in another dimension.

Computational and AI assistance. OpenAI Codex assisted in the algebraic derivations, program generation, execution, verification, and manuscript preparation. A separate model context examined the mathematical argument and recomputed the decisive values using separately authored implementations. This was a separate computational review pass, not external peer review or a claim of statistical independence. The statement has not been formalized in a proof assistant.

References

- [1] T. W. Cusick, *Affine equivalence of cubic homogeneous rotation symmetric Boolean functions*, Information Sciences **181** (2011), 5067–5083. doi:10.1016/j.ins.2011.07.002. Version inspected: arXiv:1007.1938v2, October 24, 2011, especially Remark 3.9.
- [2] Zijian Zeng, *Certified Bounds for the Number of Unlabeled Posets on 17 Points*, SSRN working paper (2026). doi:10.2139/ssrn.7385118.